

MATH 215A : Algebraic Topology

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1 Homotopy Theory

Begin with a slew of definitions and attempts for each to furnish examples

Definition 1.1. We say that two maps $f_0, f_1 : X \rightarrow Y$ are homotopic, denoted $f_0 \sim f_1$, if there exists a map $F : X \times [0, 1] \rightarrow Y$ such that the restriction $f_t = F|_{X \times \{t\}}$ corresponds to f_0 when $t = 0$ and to f_1 when $t = 1$.

Intuitively we are saying that two paths are homotopic to one another if they can be continuously deformed into one another. Extending this example we can determine when two spaces are homotopic to one another.

Definition 1.2. X is homotopic to Y if there exists continuous maps $f : X \rightarrow Y : g$ such that $f \circ g \sim id_X$ and $g \circ f \sim id_Y$.

Note that if instead of homotopic we get equivalent then this is the meaning of being homeomorphic (bijective, continuous maps). In a sense we are presenting a weaker form of equivalent of two spaces. Some examples of homotopic spaces:

1. $D^n \sim pt$: Lets fix a point $c \in D^n$ which we can think of as the center of this disk. Define a map $f : D^n \rightarrow \{c\}$ to be the constant map taking everything in the disk to a point c . Define $g : \{c\} \rightarrow D^n$ to be a map which takes $g(c) = c$. Per our claim it remains to define a homotopy map $H : D^n \times [0, 1] \rightarrow D^n$ such that $H(x, 0) = f \circ g(x)$ and $H(x, 1) = id_X$. If we take the straight line homotopy

$$H(x, t) = (1 - t)c + tx$$

then we immediately satisfy those conditions. (I believe this generalizes further to the fact that any path-connected space is contractible to a point).

2. $T \sim S^1$ where T is a solid torus. Intuitively the solid torus can be seen as $D^2 \times S^1 \sim pt \times S^1$. What we have can naturally be identified with just S^1 .

Definition 1.3 (Deformation Retraction). Let $A \subset X$, we refer to the inclusion as a map i , and define $r : X \rightarrow A$. If $i \circ r = id_A$ then we call this a retraction. If $r \circ i \sim id_X$ then we call this a deformation retraction.

Primary example to consider the cylinder $X \times I$. The retraction map $r : B \times I \rightarrow B$ simply drops the interval parameter, then $r \circ i(b) = r(b \times I) = b$. We can simply define a straight line homotopy and we get the deformation retraction homotopy equivalence. Note here that $B \times I \not\cong B$ yet $B \times I \sim B$. In fact these two maps are equivalent in this case making this a *strong* deformation retraction. With this language we can precisely define what it means for a space to be contractible.

Definition 1.4. A space X is called contractible if $X \sim pt$. Alternately whenever there exists a deformation retraction to a point $x_0 \in X$ homotopic to id_X .

We already saw above an example of the n -disk being contractible. In fact if a space is path-connected then it is contractible. Now consider the set $\pi(X, Y) = E(X, Y) / \sim$ where $\sim$ is homotopy equivalence. That is we are stratifying the path space between two topological spaces into a new set where equivalence classes correspond to homotopy equivalent functions. We can check that the properties of an equivalence relation:

- Transitivity corresponds to composition of paths
- Symmetry correspond to the reversal of the direction of a path
- Reflexivity corresponds to the fact that a path is always homotopic to itself

Now we give a categorical view of this homotopy equivalence property. Fix a space Z , then $\pi(Z, -)$ and $\pi(-, Z)$ define a covariant and contravariant functor respectively from the category of topological spaces to the category of sets. That is on objects $X \mapsto \pi(Z, X)$ and $X \mapsto \pi(X, Z)$. Note that for homotopic maps $\phi \sim \phi'$ in $\pi(Z, X)$ that composition with a map $f : X \rightarrow Y$ preserves the homotopy, i.e. $f \circ \phi \sim f \circ \phi'$. Then to f we associate a map $f_*^Z : \pi(Z, X) \rightarrow \pi(Z, Y)$ where $[\phi] \mapsto [f \circ \phi]$. Similarly if we fix Z in the second coordinate we get a map $f_*^Z : \pi(Y, Z) \rightarrow \pi(X, Z)$. If we have a map $h : Z \rightarrow W$, then we get a collection of maps which behave naturally in the sense that

$$\begin{array}{ccc} \pi(W, X) & \xleftarrow{h_X^*} & \pi(Z, X) \\ \downarrow f_*^W & & \downarrow f_*^Z \\ \pi(W, Y) & \xleftarrow{h_Y^*} & \pi(Z, Y) \end{array} \quad \begin{array}{ccc} \pi(X, W) & \xrightarrow{h_*^X} & \pi(X, Z) \\ \uparrow f_*^W & & \uparrow f_*^Z \\ \pi(Y, W) & \xrightarrow{h_*^Y} & \pi(Y, Z) \end{array}$$

commutes.

Theorem 1.1. X is homotopy equivalent to Y if for all Z there exists such an $\alpha^Z : \pi(Z, X) \rightarrow \pi(Z, Y)$ natural with respect to Z .

Proof. If X and Y are homotopy equivalent then setting $\alpha^Z = f_Z^*$ suffices as the homotopy equivalence ensures that this map is bijective. Now suppose that such an α^Z exists for all Z , then the natural gives us the following commutative square:

$$\begin{array}{ccccccc} \pi(X, X) & \xrightarrow{g_X^*} & \pi(Y, X) & \xrightarrow{f_X^*} & \pi(X, X) \\ \downarrow \alpha^X & \searrow \alpha^Y g_X^* = g_Y^* \alpha^X & \downarrow \alpha^Y & \nearrow (\alpha^X)^{-1} f_Y^* = f_X^* (\alpha^Y)^{-1} & \downarrow \alpha^X \\ \pi(X, Y) & \xrightarrow{g_Y^*} & \pi(Y, Y) & \xrightarrow{f_Y^*} & \pi(X, Y) \end{array}$$

Let $\alpha^X(\text{id}_X)$ denote the homotopy equivalence class of the map $f : X \rightarrow Y$ and $(\alpha^Y)^{-1}$ denote the homotopy equivalence class of $g : Y \rightarrow X$. Then,

$$\begin{aligned} [\text{id}_X] &= \alpha^Y g_X^*(\text{id}_Y) = g_Y^* \alpha^X(\text{id}_Y) & (\text{Commutativity}) \\ &= g_Y^*([f]) & (1) \\ &= [f \circ g] & (2) \end{aligned}$$

and

$$\begin{aligned} [\text{id}_Y] &= (\alpha^X)^{-1} f_Y^*(\text{id}_X) = f_X^* (\alpha^Y)^{-1}(\text{id}_X) & (3) \\ &= f_X^*([g]) & (4) \\ &= [g \circ f] & (5) \end{aligned}$$

The equivalence of these equivalence classes implies directly that $f \circ g \sim \text{id}_Y$ and $g \circ f \sim \text{id}_X$ implying that X is homotopy equivalent to Y . 8

2 Homotopy Equivalence of Maps

Consider that any map $f : X \rightarrow Y$ factors as follows

$$X \xleftarrow{i} X \times Y \xrightarrow{\pi} Y \quad \text{with } f = \pi \circ i$$

Definition 2.1 (Homotopy Equivalence). We say $\tilde{f} : \tilde{X} \rightarrow \tilde{Y}$ is homotopy equivalent to $f : X \rightarrow Y$ if there exists maps $\phi : \tilde{X} \rightarrow X$, $\psi : \tilde{Y} \rightarrow Y$ such that $f \circ \phi \sim \psi \circ \tilde{f}$.

$$\begin{array}{ccc} \tilde{X} & \xrightarrow{\tilde{f}} & \tilde{Y} \\ \downarrow \phi & & \downarrow \psi \\ X & \xrightarrow{f} & Y \end{array}$$

Suppose that we have a map $f : X \rightarrow Y$, take $\tilde{Y} = \text{cyl}(f)$ and note that $r : \text{cyl}(f) \rightarrow Y$ is a strong deformation retraction then our square becomes

$$\begin{array}{ccc} X & \xhookrightarrow{i} & \text{cyl}(f) \\ \downarrow \text{id}_X & & \downarrow r \\ X & \xrightarrow{f} & Y \end{array}$$

If we can show that $f \circ \text{id}_X \sim r \circ i$ then we can construct a homotopy equivalence for any map f . Note that f is exactly $r \circ i$ and $\text{id}_X \circ f = f$ and of course a function is homotopic to itself and so we are done. Though we are not forced into using the mapping cylinder, as long as we can factor the map as an inclusion/projection through the object of choice then we can construct the homotopy equivalence. In such a case we ask that given some projection $\pi : E \rightarrow B$ that lives in this homotopy equivalence square, could we lift paths from B to E . Suppose that we have a homotopy $f_t : Z \times I \rightarrow B$, then we get a natural inclusion from $Z \times \{0\}$ into $Z \times I$ which we pair with a "lift". This lift is a map $\tilde{f}_0 : Z \times \{0\} \rightarrow E$ such that $\pi \circ \tilde{f}_0 = f_0$. Then we are furnished with a "covering homotopy", a family of maps $\tilde{f}_t$ such that $Z \times I \rightarrow E$ where $\pi \circ \tilde{f}_t = f_t$.

Definition 2.2. A map $\pi : E \rightarrow B$ is said to have the *Covering Homotopy Property (CHP)* if we get the square

$$\begin{array}{ccc} Z \times 0 & \xrightarrow{\tilde{f}_0} & E \\ \downarrow i & \nearrow (\tilde{f}_t) & \downarrow \pi \\ Z \times I & \xrightarrow{f_t} & B \end{array}$$

More precisely if for any homotopy f_t and a lift $\tilde{f}_0$ there exists a covering homotopy $(\tilde{f}_t)$.

Naturally, as we are in a homotopy theory class, we ask ourselves whether paths in B can be lifted to paths in E . The answer is - almost. If B is a path-connected space then the fibers of the projection $\pi^{-1}(x_i)$ are homotopy equivalent to one another.

Theorem 2.1. If a map $\pi : E \rightarrow B$ over a path connected space B has the covering homotopy property then the fibers of the projection π are homotopy equivalent.

Proof. Let $x_1, x_2 \in B$ be points connected with a path $\gamma : [0, 1] \rightarrow B$. Denote $F_i = \pi^{-1}(x_i)$ to be the fibers of the projection over elements of B . Observe that naturally $F_0 \hookrightarrow E$ and take $Z = F_0$ treating this inclusion as a lift over the homotopy $f : F_0 \times [0, 1] \rightarrow B$ where $f = \gamma \circ \text{pr}_2$ that is we project to the interval and utilize the path we have defined earlier. The CHP furnishes us with a family of $\tilde{f}_t : F_0 \rightarrow F_t$ to the fibers of F_t over $\gamma(t)$, note that the choice to map to the fibers over points of the path is natural because we need $\pi \circ \tilde{f}_t = f(z, t)$. With this we get a map $f_1 : F_0 \rightarrow F_1$. Taking $Z = F_1$ we obtain a family of maps $\tilde{g}_t : F_1 \rightarrow F_{1-t}$ in particular we will pick out a map $g_1 : F_1 \rightarrow F_0$. To show that F_1 and F_0 are homotopy equivalent we will show that f_1 and g_1 are homotopy inverse i.e. $g_1 \circ f_1 \sim \text{id}_{F_1}$ and $f_1 \circ g_1 \sim \text{id}_{F_0}$. We will utilize the CHP property of π again by take $Z = F_0 \times [-1, 1]$ with a family of maps $(h_t) : Z \rightarrow E$ such that

$$h_t(z) = \begin{cases} f_{1+t}(z) & , t \leq 0 \\ g_1 \circ f_1 & , t > 0 \end{cases}.$$

Remark 2.2. *Intuitively we are thinking of this as connecting the lifts of the forward path with the reverse path stringing this together into one homotopy path. Then we will choose another parameters that will control the shrinking of the path in B and modify it's corresponding family of paths in E .*

Observe that with this definition $h_{-1}(z) = f_0(z) = \text{id}_{F_0}$ and $h_1(z) = g_1 \circ f_1$ Consider a new homotopy $\delta_u : [-1, 1] \times [0, 1] \rightarrow B$ such that

$$\delta_u(t) = \begin{cases} \gamma(u(1+t)) & , t \leq 0 \\ \gamma(u(1-t)) & , t > 0 \end{cases}.$$

Observe that $\delta_0(t) = \gamma(0) = x_0$ for all $t \in [-1, 1]$ effectively acting as the constant path. Now a homotopy $F_0 \times [-1, 1] \xrightarrow{pr_2} [-1, 1] \xrightarrow{\delta_u} B$, via the CHP, can be lifted to a family of maps parametrized by t and u , $h_{t,u} : F_0 \times [-1, 1] \times [0, 1] \rightarrow E$ where $h_t(1) = \delta_1(t)$ for all $t \in [-1, 1]$ is equivalent to the path h_t . Now consider the path traced over segments $\{-1\} \times [0, 1]$, $[-1, 1] \times 0$, and $\{1\} \times [0, 1]$ with endpoints $(-1, 1)$ and $(1, 1)$. As the maps

$$h_{-1,1} = h_{-1} = \text{id}_{F_0} \quad \text{and} \quad h_{1,1} = h_1 = g_1 \circ f_1$$

we have successfully constructed a homotopy equivalence of $\text{id}_{F_0} \sim g_1 \circ f_1$. 8

As we established before this, for any map $f : X \rightarrow Y$ we can always construct a homotopically equivalent map $\tilde{f} : X \rightarrow \text{Cyl}(f)$ but we can actually simplify this by instead taking $E = X \times E(Y) = \{(x, \gamma) : \gamma(1) = f(x)\}$. Not only does this give us a homotopy equivalence we can further extend this to show that this has the CHP. This establishes the fact that we only need one map to construct a homotopically equivalent inclusion from a total space to Y .

Remark 2.3. *My understanding of this is that this allows us to study maps with "nicer" properties without losing topological information, up to homotopy equivalence of course.*

Observe that a map $Z \rightarrow E$ is defined utilizing a map $\phi : Z \rightarrow X$ and $\psi : Z \times I \rightarrow Y$ where $\psi(z, 0) = \phi(f(z))$. Now construct a map $\xi : Z \times [1, 2] \rightarrow Y$ where $\xi(z, 1) = \psi(z, 1)$. Observe that $\psi(z, t) = \tau_z(t)$ giving us a family of paths $\tau_z \in E(Y)$ parametrized by $z \in Z$. Then to each of these paths we are essentially just adjoining that path $\xi(z, 1)$ ranging over $[1, 1+t]$. Then it is a matter of reparametrizing the interval and we get a homotopy that lifts automatically for every t .

3 CW - Complexes

CW Complexes have two ways of being presented but before this I will give an intuitive definition. A CW-Complex is space that is constructed by gluing the Disks together so that boundaries of higher dimensional disks correspond to lower dimensional disks of this space.

Definition 3.1 (Constructive). *Let $sk_0(X)$ be a discrete union of disks D_α^0 , i.e. this is a discrete space and it's cells are disks of dimension 0. Define the n -skeleton to be*

$$sk_n(X) = sk_{n-1}(X) \bigcup_{\alpha} D_\alpha^n$$

where the union with the α is signifying that these spaces are joined via an attaching map $\phi_\alpha : \partial D_\alpha^n \rightarrow sk_{n-1}(X)$. Similarly we can express the n skeleton as a quotient

$$sk_{n-1}(X) \bigsqcup_{\alpha} D_\alpha^n / \sim$$

where $x \in \partial D_\alpha^n$ is identified with $\phi_\alpha(x) \in sk_{n-1}(X)$.

$sk_n(X)$ is frequently denoted X^n , and with this construction we obtain a nested sequence of inclusion $X^0 \subset X^1 \subset \dots \subset X^n \subset \dots$. X is their union equipped with a direct limit topology.

Definition 3.2 (Set-Theoretic). Take X to be the union of all cells $\overset{\circ}{D}_\alpha^n$ and for each cell we can define the characteristic map

$$\Phi_\alpha : D_\alpha^n \rightarrow X$$

where ∂D_α^n coincides with the ϕ_α .

Similar to direct limit topologies, a subset in X is closed, open, if and if its inverse image in all cells D_α^n is closed, open.

Definition 3.3 (Axiomatic). A space X is equipped with a CW-structure if it is partitioned into pairwise disjoint subsets e_α^n provided with the characteristic maps $\Phi_\alpha : D_\alpha^n \rightarrow X$ which supply the homeomorphism $e_\alpha^n \simeq \overset{\circ}{D}_\alpha^n$ satisfying the following conditions:

1. *Closure-Finiteness:* Each $\Phi_\alpha|_{\partial D_\alpha^n}$ lands in finitely many cells of lower dimensions
2. *Weak Topology* A subset in X is closed iff its inverse image under each characteristic map is closed.

At this point it's a good idea to give a good example. As the nested inclusions hint, we should be able to construct the $*^\infty$ spaces using skeletons. First we will construct a finite space, S^2 . Let $X_0 = \{p\}$ be a single point, then take $X_1 = X_0$, and finally we can take $X_2 = D^2$ with a constant attaching map $\phi : D_2 \rightarrow \{p\}$ be the constant map mapping every point to p . Of course this means that the map behaves as we expect on the boundary of D^2 . Note that there multiple ways to construct a CW-complex for S^2 .